

Data-Driven Customer Analytics for E-Commerce Decision Making

An Integrated Statistical and Machine-Learning Approach to Customer Value, Churn Risk, Segmentation, and Marketing Optimization

Project 2

Customer Analytics for E-Commerce

Author

Mahdiyeh Mirzaei

Team / Community

StudyBuild Community

github.com/mahdiyeh-mirzaei-v2

github.com/StudyBuildCommunity

Abstract

This study presents an integrated customer analytics framework for an e-commerce company using statistical analysis, unsupervised learning, and supervised machine learning. The analytical dataset contains 60 customers and 26 original variables covering demographic, transactional, behavioral, and satisfaction-related characteristics. The methodology combines exploratory data analysis, descriptive statistics, normality testing, Interquartile Range (IQR)-based outlier detection, Spearman rank correlation, customer value scoring, K-Means clustering, non-parametric hypothesis testing, churn-risk classification, and predictive modeling.

The results indicate substantial heterogeneity in customer economic value. Customers were divided into three equally sized value groups, including 20 High-Value, 20 Medium-Value, and 20 Low-Value customers. A total of 36 customers, corresponding to 60% of the sample, were classified as churn-risk customers using an inactivity threshold of more than 180 days. Importantly, 12 customers were simultaneously classified as High Value and Churn Risk, identifying a strategically critical retention group.

K-Means clustering identified three behavioral segments. The most valuable cluster consisted of only six customers but exhibited an average total spending of approximately 14,188.76. The Kruskal–Wallis test confirmed statistically significant differences in spending distributions across clusters ($H = 16.109$, $p = 0.000318$).

The discount analysis did not identify a statistically significant difference in spending between customers using discounts and those not using discounts. The Mann–Whitney U test resulted in $U = 391$ and $p = 0.4512$. The machine-learning analysis also produced limited predictive performance, with mean cross-validated ROC-AUC values of approximately 0.564 for Logistic Regression and 0.478 for Random Forest.

At the geographical level, Mashhad generated the highest observed total revenue of approximately 64,118.44 and obtained the highest marketing opportunity score. Based on these findings, the study recommends prioritizing high-value churn-risk customers, increasing average order value among frequent low-basket customers, and validating marketing investment in Mashhad through controlled experimentation.

Keywords: Customer Analytics; E-Commerce; Customer Value; Churn Risk; K-Means; Logistic Regression; Random Forest; Spearman Correlation; Non-Parametric Statistics; Marketing Analytics.

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1. Introduction

The rapid expansion of e-commerce has created large volumes of customer-level data containing information about purchasing behavior, transaction value, customer satisfaction, discounts, returns, and demographic characteristics. The availability of such data provides organizations with an opportunity to move from intuition-based marketing toward evidence-based decision making.

However, raw customer data do not directly provide managerial insight. Organizations must identify economically valuable customers, recognize inactive or potentially lost customers, understand heterogeneous behavioral patterns, and determine where marketing resources are most likely to generate incremental value.

The central research question of this project is:

How can customer-level behavioral and transactional data be transformed into statistically supported and actionable strategies for improving customer retention, revenue generation, and marketing efficiency?

The analysis addresses five principal managerial questions:

1. Which customers generate the greatest economic value?
2. Which customers are potentially at risk of becoming inactive?
3. Which customer segments exhibit distinct purchasing behavior?
4. Which geographical market provides the strongest marketing opportunity?
5. Is discount usage associated with higher customer spending?

In addition, supervised machine-learning models are evaluated to determine whether customer characteristics provide useful information for identifying churn-risk customers.

The purpose of this study is therefore not merely descriptive. The analytical framework explicitly connects statistical evidence to business decisions.

2. Data Description and Preparation

2.1. Dataset Overview

The final analytical dataset contains 60 customers and 26 original variables. The variables represent demographic, transactional, behavioral, and customer-satisfaction characteristics.

Important variables include:

- `customer_id`
- `age`
- `city`

- province
- membership_tier
- purchase_count
- avg_order_value
- total_spending
- last_purchase_days
- discount_used
- returned_items
- satisfaction_score

Several analytical variables were subsequently engineered, including:

- Customer Value Score
- Customer Value Segment
- Churn Risk
- Churn Risk Label
- K-Means Cluster

2.2. Data Quality

The data-quality audit identified no duplicate observations. Therefore,

$$N = 60, \quad \text{Duplicate Rows} = 0.$$

Table 1 summarizes the main properties of the analytical dataset.

Table 1: Summary of the analytical dataset

Property	Value
Customers	60
Original variables	26
Final analytical variables	31
Duplicate rows	0
Active customers	24
Churn-risk customers	36
High-value customers	20
Medium-value customers	20
Low-value customers	20
K-Means clusters	3

3. Analytical Framework and Methodology

The complete analytical workflow is illustrated in Figure 1.

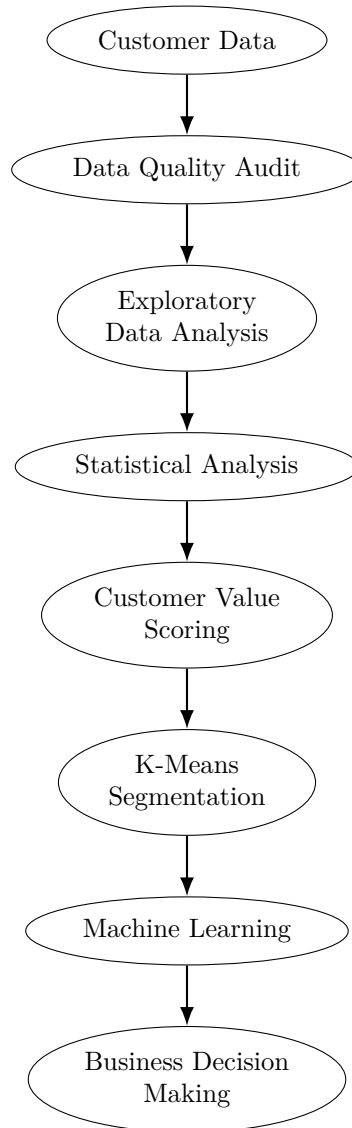


Figure 1: Integrated analytical framework used in the study

The workflow combines descriptive, inferential, unsupervised, and supervised analytical techniques.

The main Python libraries used in the project include:

- pandas
- NumPy
- SciPy
- scikit-learn
- matplotlib
- seaborn

- openpyxl
- joblib

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to characterize the distributions of customer age, purchase frequency, average order value, total spending, inactivity, and satisfaction.

Because the sample contains only 60 observations, numerical summaries were considered together with distributional diagnostics and graphical inspection.

The main numerical variables used throughout the analysis were:

- Age
- Purchase Count
- Average Order Value
- Total Spending
- Last Purchase Days
- Satisfaction Score

The distributions indicate substantial heterogeneity, particularly in total spending and purchase behavior.

5. Normality and Outlier Analysis

5.1. Normality Testing

The Shapiro–Wilk test was applied to assess the normality of the main numerical variables.

The null hypothesis was:

$$H_0 : X \sim \text{Normal Distribution.}$$

The alternative hypothesis was:

$$H_1 : X \not\sim \text{Normal Distribution.}$$

The results are presented in Table 2.

Table 2: Shapiro–Wilk normality test

Variable	W	p-value
Age	0.938	0.004
Purchase Count	0.957	0.033
Average Order Value	0.915	< 0.001
Total Spending	0.708	< 0.001
Last Purchase Days	0.970	0.142
Satisfaction Score	0.879	< 0.001

At the 5% significance level, most variables reject the normality hypothesis. Consequently, non-parametric statistical procedures are preferred for inferential comparisons.

5.2. Outlier Detection

Outliers were investigated using the Interquartile Range (IQR) criterion:

$$IQR = Q_3 - Q_1.$$

The lower and upper bounds are defined as:

$$L = Q_1 - 1.5(IQR),$$

$$U = Q_3 + 1.5(IQR).$$

Observations outside $[L, U]$ are potential statistical outliers.

High-spending observations were not automatically removed because they may represent genuine economically important customers.

For example, Customer 1030 has:

$$\text{Total Spending} = 25,000, \tag{1}$$

$$\text{Purchase Count} = 26, \tag{2}$$

$$\text{Last Purchase Days} = 340. \tag{3}$$

This observation is statistically unusual but potentially highly valuable from a managerial perspective.

6. Correlation Analysis

Because several variables exhibited non-normal distributions, Spearman's rank correlation coefficient was used.

Spearman's coefficient is defined as:

$$\rho_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)},$$

where d_i represents the difference between the ranks of the two variables for observation i .

The strongest relationships are summarized in Table 3.

Table 3: Selected Spearman rank correlations

Variable Pair	ρ_s
Purchase Count – Total Spending	0.613
Average Order Value – Total Spending	0.434
Last Purchase Days – Satisfaction	-0.293
Age – Purchase Count	0.265

The strongest association was observed between Purchase Count and Total Spending:

$$\rho_s = 0.613.$$

This indicates a moderate positive monotonic relationship, suggesting that customers who purchase more frequently tend to generate greater cumulative spending.

Nevertheless, correlation does not imply causation.

7. Customer Value Analysis

A composite Customer Value Score was developed using standardized versions of Total Spending, Purchase Count, and Average Order Value.

The score is defined as:

$$CVS = 0.50Z(TS) + 0.30Z(PC) + 0.20Z(AOV),$$

where:

- TS = Total Spending,
- PC = Purchase Count,
- AOV = Average Order Value.

Total Spending receives the largest weight because it represents realized customer revenue.

Customers were divided into three equal-sized value groups using empirical quantiles.

Table 4: Customer value segments

Segment	Customers	Share
Low Value	20	33.3%
Medium Value	20	33.3%
High Value	20	33.3%

The segmentation provides a transparent basis for differentiated customer management.

8. Churn-Risk Analysis

Churn risk was operationalized using customer inactivity.

A customer was classified as churn-risk when:

$$\text{Last Purchase Days} > 180.$$

The resulting classification is presented in Table 5.

Table 5: Churn-risk classification

Status	Customers	Percentage
Active	24	40%
Churn Risk	36	60%

Thus,

$$\frac{36}{60} \times 100 = 60\%.$$

A particularly important group consists of customers who are both High Value and Churn Risk.

There are 12 such customers:

$$\frac{12}{60} \times 100 = 20\%.$$

This group should receive the highest retention priority.

It is important to emphasize that churn is defined here as an inactivity-based proxy rather than an observed future churn outcome.

9. Customer Segmentation Using K-Means

K-Means clustering was applied using:

- Purchase Count,
- Average Order Value,
- Total Spending,
- Last Purchase Days.

The variables were standardized before clustering.

The quality of candidate cluster solutions was assessed using the silhouette coefficient.

Table 6: Silhouette scores for candidate cluster solutions

Number of Clusters	Silhouette Score
2	0.280
3	0.316
4	0.297
5	0.308
6	0.299

The highest silhouette score occurred at $K = 3$.

Therefore, three clusters were selected.

10. Cluster Profiles and Statistical Validation

The three identified clusters are summarized in Table 7.

Table 7: K-Means customer cluster profiles

Cluster	Customers	Avg. Purchases	Avg. AOV	Avg. Spending	Avg. Inactivity
Cluster 0	6	30.33	348.60	14,188.76	211.17
Cluster 1	23	8.30	309.07	2,603.76	232.17
Cluster 2	31	21.61	115.78	2,475.26	173.81

10.1. Cluster 0: Premium Customers

Cluster 0 contains only six customers but has an average total spending of approximately 14,188.76.

This cluster represents the company’s most economically valuable behavioral group.

Recommended strategies include VIP retention, personalized communication, premium services, early access, and targeted relationship management.

10.2. Cluster 1: Low-Frequency Customers

Cluster 1 contains 23 customers with an average purchase count of 8.30 and average inactivity of approximately 232 days.

This group is suitable for reactivation strategies and personalized engagement.

10.3. Cluster 2: Frequent Low-Basket Customers

Cluster 2 contains 31 customers with relatively high purchase frequency but an average order value of only 115.78.

This segment provides a strong opportunity for cross-selling, bundling, and upselling.

10.4. Kruskal–Wallis Validation

To statistically evaluate whether total spending differs across clusters, the Kruskal–Wallis test was applied.

The test statistic was:

$$H = 16.109,$$

with:

$$p = 0.000318.$$

Since:

$$p < 0.05,$$

the null hypothesis of identical spending distributions across the clusters is rejected.

Therefore, statistically significant differences in customer spending exist across the identified segments.

11. Machine-Learning Churn Analysis

Two supervised learning algorithms were evaluated:

1. Logistic Regression
2. Random Forest

Predictors included:

- Age
- Purchase Count
- Average Order Value
- Total Spending

- Satisfaction Score
- Returned Items

The variable `last_purchase_days` was excluded because it directly defines the churn target. Including it would introduce target leakage.

11.1. Cross-Validation Results

Stratified five-fold cross-validation was used.

Table 8: Cross-validated machine-learning performance

Model	Mean ROC-AUC	Std. Dev.
Logistic Regression	0.564	0.084
Random Forest	0.478	0.096

The results indicate weak predictive performance.

A ROC-AUC value near 0.50 corresponds approximately to random classification. Logistic Regression performs somewhat better than Random Forest, but neither model is sufficiently strong for production deployment.

12. Random Forest Feature Importance

The estimated Random Forest feature importances are presented in Table 9.

Table 9: Random Forest feature importance

Feature	Importance
Total Spending	0.235
Average Order Value	0.214
Satisfaction Score	0.167
Age	0.156
Purchase Count	0.149
Returned Items	0.078

Total Spending and Average Order Value were the two most influential variables within the fitted Random Forest.

These importance values should not be interpreted as causal effects.

13. Discount Effectiveness Analysis

Customers were divided into discount users and non-users.

Table 10: Descriptive comparison of discount groups

Metric	Discount	No Discount
Customers	26	34
Mean Total Spending	3,642.42	3,736.74
Mean AOV	203.29	220.70
Mean Satisfaction	3.27	2.76

Because the data are non-normal, the Mann–Whitney U test was applied.

The results were:

$$U = 391,$$

$$p = 0.4512.$$

Since:

$$p > 0.05,$$

there is no statistically significant evidence of a difference in total spending between the two groups.

The estimated Cohen’s d was approximately:

$$d = -0.022.$$

The bootstrap 95% confidence interval for the mean spending difference was approximately:

$$[-2146.19, 2400.88].$$

Because the confidence interval contains zero, the data do not provide strong evidence of a systematic spending difference.

Importantly, this does not demonstrate that discounts are ineffective. The observational design does not establish causality.

14. City-Level Marketing Opportunity

City-level marketing opportunity was evaluated using customer count, total revenue, average customer value, and satisfaction.

Table 11 presents the principal city-level results.

Table 11: Selected city-level customer and revenue indicators

City	Customers	Total Revenue	Avg. Spending	Avg. Satisfaction
Mashhad	11	64,118.44	5,828.95	2.91
Tabriz	12	36,192.49	3,016.04	3.00
Karaj	7	26,189.81	3,741.40	2.86
Ahvaz	8	23,384.78	2,923.10	3.50
Isfahan	5	21,975.32	4,395.06	4.40
Shiraz	6	20,715.92	3,452.65	2.00
Rasht	5	15,168.54	3,033.71	2.60
Tehran	6	14,006.85	2,334.48	2.67

Mashhad achieved the highest observed total revenue:

$$R_{\text{Mashhad}} = 64,118.44.$$

It also recorded the highest average customer spending among the listed cities.

Therefore, Mashhad represents the strongest candidate for a controlled marketing experiment.

However, the average satisfaction score of 2.91 indicates that customer-experience problems should be investigated before aggressively scaling acquisition.

15. Business Recommendations

The analytical results lead to three principal recommendations.

15.1. Recommendation 1: High-Value Customer Retention

The company should immediately prioritize the 12 customers who are both High Value and Churn Risk.

Potential actions include:

- Personalized communication
- VIP benefits
- Exclusive product recommendations
- Early product access
- Personalized incentives
- Customer-service follow-up

The main performance indicators should include:

- Reactivation Rate
- Revenue Recovered
- Repeat Purchase Rate
- Customer Lifetime Value
- Cost per Reactivated Customer

15.2. Recommendation 2: Increase Average Order Value

The 31 customers in Cluster 2 represent a major upselling opportunity.

Recommended tactics include:

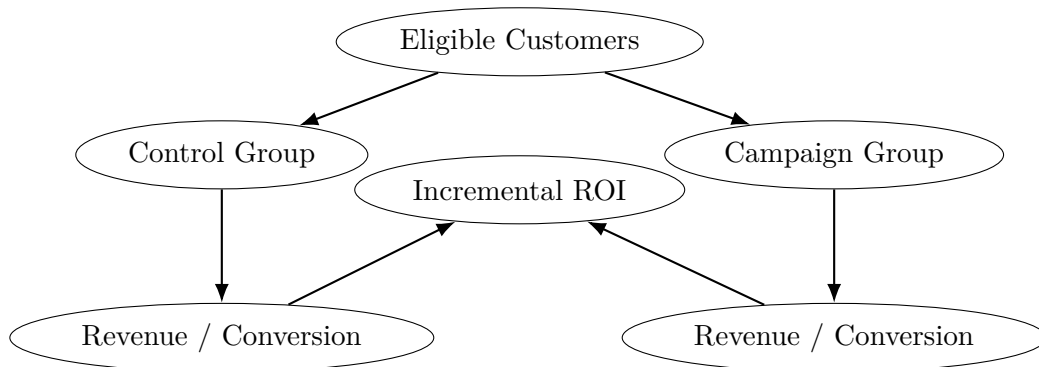
- Product bundles
- Cross-selling
- Complementary-product recommendations
- Free-shipping thresholds
- Quantity-based offers
- Premium-product recommendations

The primary KPI should be Average Order Value.

15.3. Recommendation 3: Controlled Marketing Experiment in Mashhad

Mashhad should be considered for a controlled marketing pilot.

A randomized experimental design is recommended:



The experiment should evaluate incremental revenue, conversion, repeat purchasing, customer satisfaction, and Return on Advertising Spend (ROAS).

16. Customer Strategy Matrix

Table 12: Recommended customer-management strategies

Customer Type			Main Problem	Recommended Action
High Value	+	Churn Risk	Risk of losing high revenue	Immediate personalized retention
High Value	+	Active	Protect existing value	VIP loyalty and relationship management
Frequent AOV	+	Low	Low basket size	Upselling and cross-selling
Low Frequency	+	Churn Risk	Low engagement	Reactivation campaign
Discount Users			Unclear incremental value	Targeted A/B-tested discounts
Mashhad customers		Cus-	High commercial opportunity	Controlled marketing experiment

17. Limitations

17.1. Small Sample Size

The dataset contains only 60 customers. Consequently, machine-learning estimates may be unstable and should be considered exploratory.

17.2. Proxy Churn Definition

Churn was defined using:

$$\text{Last Purchase Days} > 180.$$

This is a business-rule proxy rather than an observed future churn event.

17.3. Observational Discount Analysis

Customers were not randomly assigned to discount and non-discount groups. Therefore, causal conclusions about discount effectiveness cannot be established.

17.4. Potential Sampling Bias

The available customer sample may not represent the complete customer population.

17.5. Machine-Learning Performance

The Logistic Regression model achieved a mean ROC-AUC of approximately 0.564, while Random Forest achieved approximately 0.478.

These results do not justify production deployment.

18. Future Research

A production-level customer analytics system should incorporate longitudinal transaction data.

Important additional variables include:

- Transaction timestamps
- Product identifiers
- Product categories
- Quantity
- Price
- Discount amount
- Gross revenue
- Profit margin
- Website sessions
- App activity
- Cart abandonment
- Email engagement
- Campaign exposure

A genuine churn target should be defined as a future behavioral outcome.

For example:

$$Y_i = \begin{cases} 1, & \text{if customer } i \text{ does not purchase during the future window,} \\ 0, & \text{otherwise.} \end{cases}$$

A temporal train-test split should then be used to estimate true out-of-time predictive performance.

Potential future algorithms include:

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost

- Survival Analysis

Performance should be evaluated using ROC-AUC, Precision-Recall AUC (PR-AUC), calibration, and business-cost metrics.

19. Conclusion

This study demonstrates how customer-level data can be transformed into actionable business intelligence through an integrated statistical and machine-learning framework.

The analysis revealed substantial heterogeneity in customer value. Twenty customers were classified as High Value, while K-Means clustering identified a particularly valuable behavioral segment containing only six customers with an average total spending of approximately 14,188.76.

The churn-risk analysis identified 36 inactive customers, representing 60% of the sample under the selected inactivity rule. More importantly, 12 customers were simultaneously classified as High Value and Churn Risk. These customers represent the most strategically important retention group.

The K-Means segmentation was supported by the Kruskal–Wallis test, which produced:

$$H = 16.109, \quad p = 0.000318.$$

Thus, customer spending differs significantly across the identified behavioral segments.

The discount analysis did not detect a statistically significant spending difference:

$$U = 391, \quad p = 0.4512.$$

Consequently, broad discounting cannot be justified based solely on the current observational evidence. Controlled experiments are recommended to estimate the incremental effect of discounts.

At the geographical level, Mashhad achieved the highest observed total revenue of approximately 64,118.44. However, its moderate satisfaction level indicates that marketing expansion should be accompanied by improvements in customer experience.

The machine-learning results were limited. Logistic Regression achieved a mean cross-validated ROC-AUC of approximately 0.564, whereas Random Forest achieved 0.478. These findings indicate that the current dataset is not sufficiently large or longitudinal for reliable production churn prediction.

Overall, the evidence supports a differentiated customer strategy rather than a one-size-fits-all marketing approach.

The recommended priority is therefore:

Protect High-Value Customers → Increase AOV → Validate Marketing Through Experiments

This framework provides a practical foundation for future customer analytics systems once larger

and longitudinal datasets become available.

20. Final Management Action Plan

Table 13: Prioritized management action plan

Priority	Action	Target	Primary KPI
1	High-value retention	12 High Value + Churn Risk	Reactivation Rate
2	Upselling and cross-selling	31 frequent low-AOV customers	Average Order Value
3	Marketing pilot	Mashhad	Incremental Revenue / ROI
4	Discount A/B testing	Selected customer groups	Revenue Lift
5	Improve churn data	All customers	Future ROC-AUC / PR-AUC

The principal managerial conclusion is that customer resources should be allocated according to economic value, behavioral characteristics, and risk rather than treating all customers identically.

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